

Erdős Problem #890

EP-890

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

If $\omega(n)$ counts the number of distinct prime factors of n , then is it true that, for every $k \geq 1$, $\liminf_{n \rightarrow \infty} \sum_{0 \leq i < k} \omega(n+i) \leq k + \pi(k)$? Is it true that $\limsup_{n \rightarrow \infty} \left(\sum_{0 \leq i < k} \omega(n+i) \right) \frac{\log \log n}{\log n} = 1$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>